

1. Differentiate between ethics, law and professional responsibility with suitable examples from the computing field.

Ethics, law and professional responsibility are related but different ways of guiding behaviour in computing. They differ in source of authority, enforcement and scope of obligation.

Ethics:

Ethics refers to moral principles that guide what is wrong or right behaviour. It is based on values, fairness and integrity - not necessarily written or enforced by the government.

Computing example

A software developer avoids designing a feature that manipulates a user to share more data than they need, even though it is legal. The developer believes it is unfair and deceptive.

Ethics often goes beyond legal requirements.

Law:

Law consists of rules and regulations created and enforced by a government. Violating laws result in legal penalties such as fines or imprisonment. It defines the minimum acceptable behaviour required by society.

Computing example:

A company collects personal data user consent and violates data protection regulation. The company may face legal penalties.

Professional responsibility:

Professional responsibility refers to the duties and standards expected from members of a profession. It combines ethical behaviour with technical competence.

Computing example:

A software engineer properly tests a system before release to prevent harm to users.

2. Discuss any two historical cases of ethical failures in technology and explain the lessons learned from them.

i) Facebook - Cambridge Analytica Data Scandal (2018)

A political consulting firm, Cambridge Analytica, called collected personal data of millions of facebook users without proper consent through a quiz app. The data was used to influence political campaigns and voter behaviour.

Ethical issue:

- i) Violation of users privacy.
- ii) Lack of informed consent.
- iii) Misuse of personal data.

Lessons learned:

- i) Companies must be transparent about data collection.
- ii) Strong data protection law are useful.

iii) Users should have control over their data.

ii) Volkswagen Emissions Software Scandal (2015)

Volkswagen installed software in diesel cars to cheat emission tests. The cars detected when they were being tested and reduced emissions temporarily, but produced higher pollution in real driving.

Ethical issues:

- i) Internal deception using software.
- ii) Ignoring environment and public health.
- iii) Corporate pressure overriding ethical engineering.

Lesson learned:

- i) Engineers must prioritize public safety.
- ii) Strong regulation oversight is necessary.
- iii) Ethical culture in organizations prevents misconduct.

3. Explain the principles of software engineering ethics, highlighting the issues related to professional responsibility. Discuss the ACM/IEEE code of ethics guides ethical decision making in software engineering practices.

Principle of Software Engineering Ethics

1. **Public interest:** Software engineers must prioritize the safety, well-being and rights of the public. Systems should not cause harm or risk to others.
2. **Quality and reliability:** Engineers must produce high quality, secure and reliable software through proper design and testing and maintenance.
3. **Honesty and Integrity:** Engineers must be truthful about system capabilities, risks and limitations.
4. **Confidentiality and Privacy:** Sensitive data must be protected and user privacy must be respected.
5. **Fairness and Non-discrimination:** Software should not create unfair bias or discrimination.

6. Professional competence Engineers should work only within their expertise and improve skills.

Issues related to professional responsibility

1. Ensuring user safety
2. Protecting confidential info of clients and users.
3. Reporting risk or system failures.
4. Avoiding conflict of interest, such as promoting unsafe products.
5. Taking accountability for errors of software.

ACM/IEEE code of ethics:

The ACM/IEEE Software engineering Code of Ethics provides a structured framework that helps software engineers make responsible and morally sound decisions in complex professional situations.

How ACM/IEEE code of ethics guides practice:

- Provides clear standards for responsible behaviour in design, development and management of software.
- Help resolve ethical dilemmas.
- Encourages accountability.
- Promotes professionalism.

4. Compare and contrast Utilitarianism and Deontology as ethical frameworks. How would each approach handle a software privacy dilemma?

Utilitarianism

It is a consequence based theory. An action is morally right if it produces the greatest happiness for the greatest number of people.

Deontology

Deontology is a duty based theory. An action is morally right if it follows moral rules or duties, regardless of consequences.

Comparison between Utilitarianism and Deontology:

Basis of comparison	Utilitarianism	Deontology
Founders	Jeremy Bentham	Immanuel Kant
Type of theory	Consequence based	Duty based
Main focus	Outcomes of actions	Moral duties and rules
Moral principle	Greatest happiness for the greatest number	Act according to universal moral law
Role of consequence	Very important	Not important
Decision Criterion	An action is right if it maximizes overall utility	An action is right if it follows moral duty
Flexibility	Flexible & situational	Rigid and rule based
Example	Allow data collection if it benefits majority of users	Reject data collection without consent.

Strength	Promotes social welfare	Ensures fairness and respect for persons.
Weakness	can justify injustice	can ignore beneficial outcomes.

In a software privacy dilemma, a company wants to collect user's personal data without full consent to improve services or security.

Utilitarianism approach:

Utilitarianism focuses on overall consequences and the greatest good for greatest number.

- If collecting user data benefits many people, it may be considered morally acceptable.
- privacy loss for some individuals may be justified if total benefits are greater than total harm.

Deontology approach:

Deontology focuses on moral duties or individual rights, not on consequences.

- Users have right to privacy and informed consent.

- collecting data without permission violates moral duty and treats people as a means to an end.

5. Why is explainability a key ethical requirement for AI systems used in high-stakes domains, like credit lending or criminal justice?

Discuss two societal harms that can occur if an AI system operates as a black box.

Explainability means that an AI system's decisions can be understood and justified by humans.

It is a key ethical requirement in high stakes domains.

Why explainability is ethically important

1. Fairness and accountability

People must know why a loan was denied or a risk score assigned. Without explanations, unfair or biased decisions can't be corrected.

2. Trust and transparency

Institutions must be able to justify decisions to the public, regulators and affected individuals.

3. Error detection and correction

If decisions are explainable, experts can identify mistakes, bias or faulty data and fix them.

Two societal harms of black-box AI

1. Discrimination and unfair treatment

If the system's reasoning is hidden, it may produce biased decisions based on race, gender, income or location.

People could be denied loans or treated more harshly in legal decisions without knowing why.

2. Lack of accountability and injustice

When no one understands how decisions are made, responsibility becomes unclear. Wrong or harmful decisions can't be properly challenged, corrected or legally reviewed.

6. What is whistleblowing? Discuss the ethical considerations and challenges a computing professional might face when deciding to blow the whistle.

Whistleblowing: Whistleblowing is the act of reporting illegal, unethical or harmful activities within an organization to authorities management or the public to prevent wrongdoing and protect society.

Ethical considerations

1. Public safety and harm prevention:
If the issue could harm users, society or the environment, reporting it may be a moral duty.

2. Loyalty and responsibility

The professional must balance loyalty to their employer with responsibility to the public and ethical principles.

3. Evidence and truthfulness

The concern should be based on clear evidence, not assumptions.

4. Last resort principle

Ethical practice often suggests reporting internally first before going public, if possible.

Challenges faced by whistleblowers

1. Risk of retaliation

They may face job loss, demotion or legal threats.

2. Career and financial consequences

Future employment opportunities may be affected.

3. Emotional and social pressure

stress, isolation and conflict with colleagues.

4. Legal uncertainty

Whistleblowers may not always be fully protected by law.

7. Explain the difference between copyright, patents and trademarks in the context of software and digital products.

Copyright, patents and trademarks are three different types of intellectual property protection used in software and digital products.

Copyright

- Copyright protects the expression of ideas.
- In context of software, it protects the source code, object code, graphics, website content and documentation.
- It doesn't protect the underlying idea, only how it is expressed.
- Copyright protection usually arises automatically once the work is created.
- Example: copying someone else's source code without permission is copyright infringement.

Patent

- Patents protect new and innovative inventions.
- In software, a patent may protect a unique algo, data processing method or system architecture.
- Patent protection requires formal application and approval from the government.
- Unlike copyright, a patent protects the functional idea or method itself.

Trademark

- Trademark protects brand identity.
- In software, it protects the product name, logo or symbol that distinguishes one company's product from another.
- Prevents confusion by ensuring others can't use similar branding.
- Ex: A software product's name or logo is protected by trademark.

8. compare open source licensing with proprietary software licensing. What are the ethical implications of each model?

Comparison between open source Licensing and Proprietary Software Licensing:

Open Source Licensing

Open source software allows users to view, modify and distribute the source code under specific license terms.

Examples of open source licenses are issued by organization like the Open Source Initiative.

Key features:

- Source code is publicly available
- Users can modify and redistribute the software.
- Encourages transparency.
- Usually free to use.

Example

Linux operating system, Apache webserver.

Proprietary Software Licensing

Proprietary software restricts access to the source code. Users only receive permission to use the software under certain conditions.

Key features

- Source code is closed and confidential.
- Modification and redistribution aren't allowed.
- Users must purchase or obtain a license.
- Full control remains with the company.

Example

Microsoft windows, adobe photoshop

Major differences

- Open source allows modification, proprietary doesn't.
- Open source promotes collaboration, proprietary emphasizes ownership and control.

- Open source is community driven; Proprietary is company controlled.
- Open source software is free, Proprietary requires payment.

Ethical implications

Open source Licensing

- promotes transparency and trust.
- Encourages collaboration and innovation.
- Supports equal access to technology.
- May raise concerns about sustainability and fair compensation for developers.

Proprietary software Licensing

- Protects intellectual property and developer investment.
- Enables quality control and accountability by the owner.
- Limits user freedom and transparency.
- Can create restricted access or inequality due to licensing costs.

9. Discuss the concept of informed consent in personal data collection. What ethical principles should organisations follow when collecting user data?

Informed consent in personal data collection means that individuals clearly understand what personal data is being collected, why it is collected, how it will be used and who it will be shared with - and they freely agree to it before any data is gathered. It ensures respect for individual autonomy.

Key elements of informed consent:

- Clear and understandable information are given.
- Know what data is being collected.
- Understand why data is being collected.
- Are aware of being how long it will be stored.
- Know who it will be shared with.
- Have the option to accept or refuse.

Ethical principles organizations should follow:

1. **Transparency:** Organizations must clearly explain what data they collect and how it will be used.
2. **Privacy and confidentiality:** personal info must be protected from unauthorized access, misuse or disclosure.
3. **Purpose limitation:** Data should be collected only for specific, legitimate purpose and not used for other purposes without permission.
4. **Data minimisation:** Only collects data that is necessary.
5. **User control and autonomy:** Users should be able to access, correct, delete their data and withdraw consent whenever they want.
6. **Security and protection:** Strong technical measures must be used to protect data from loss.
7. **Accountability:** Organisations must take responsibility for how they collect, use and

protect data.

10. What is Algorithmic bias? Explain its causes and suggest measures to ensure fairness in AI systems.

Algorithmic Bias:

Algorithmic bias refers to systematic and unfair discrimination produced by computer algorithms, especially in AI and machine learning systems. It occurs when an AI system produces results that are prejudiced toward certain groups based on race, gender, age, location.

In simple terms, algorithmic bias happens when an AI system makes unfair decisions because of biased data.

Cause of Algorithmic Bias:

1. Biased Training Data: If the dataset used to train an AI system contains historical discrimination of certain groups, the system may learn and repeat those biases.

2. **Sampling Bias**: If certain groups are under-represented in the dataset, predictions will be less accurate for them.

3. **Human design choices**: Developers assumptions design can introduce bias.

4. **Historical and Social Inequalities**: AI system trained on real-world data may affect existing social discrimination.

5. **Lack of transparency**: "Black-box" AI models make it difficult to detect unfair decision making.

Measures to Ensure Fairness in AI systems

1. Use diverse and Representative data of different genders, ethnicities and social groups.

2. Bias Testing and auditing

3. Transparency and explainability

4. Ethical Guidelines and standards.

5. Regular monitoring and updates.

6. Human oversight: Include human review in high stakes decisions.

11. Discuss the importance of transparency and explainability in AI. Who should be held accountable when an autonomous system causes harm?

Importance of Transparency and Explainability in AI:

Why Transparency is important:

- Builds public trust.
- Helps detect errors.
- Enables regulatory compliance.
- Protects user rights.

Without transparency, AI systems become "Black boxes".

Why Explainability is important

- Supports fairness and non-discrimination.
- Allows users to challenge incorrect decisions.
- Improves system debugging and improvement.
- Ensures ethical and legal compliance.

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Accountability in Autonomous system:

1. **Developers and Engineers:** If harm results from poor design, negligence or failure to test adequately, developers share responsibility.
2. **Companies and Organizations:** The company deploying the system is generally responsible for ensuring safety and compliance.
3. **Users or operators:** If ignoring guideline causes harm, users may be partially responsible.
4. **Manufacturers or vendors:** Responsible if hardware or software defects cause failure.

12. Differentiate between ethical hacking and cybercrime. Under what conditions is hacking considered ethical?

Ethical hacking: It refers to legally authorized attempts to test and improve the security of computer systems by identifying loop-holes.

Cybercrime: Cybercrime involves illegal activities carried out using computers to harm individuals, organizations or governments.

Difference between Ethical hacking and cybercrime

Aspect	Ethical hacking	Cyber crime
Purpose	To find and fix security weakness	To exploit systems for personal gain
Permission	Done with authorization from system owner.	Done without permission
Intention	Protects system	Share data, damage systems
Legal status	Legal and approved activity	Illegal and punished by law
Outcome	Strengthen security	Causes harm or loss.

Why is hacking considered ethical:

1. **Proper authorization:** The hacker must have written permission from the owner before testing.
2. **Legal compliance:** The activity must follow national and international cyber laws.
3. **Clear scope and agreement:** Testing must stay within agreed boundaries.
4. **Responsible disclosure:** Discover vulnerabilities must be reported privately to the organization and not exposed publicly without consent.
5. **No malicious intent:** The purpose must be improve security, not to cause

harm